Spectral Phase Transitions in the Fisher Information Matrix:

Universality of the Degeneracy Swap and Classification of Transition Order

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Abstract

We construct the Fisher Information Matrix (FIM) from spin–spin correlation functions on lattice systems undergoing phase transitions and show that the singular value (SV) spectrum undergoes a *spectral phase transition*—a degeneracy swap—that encodes both the presence and the order of the underlying thermodynamic transition. On two systems with continuous (second-order) transitions—the 2D Ising model ($\nu = 1$) and the 3D Ising model ($\nu \approx 0.630$)—the lattice-direction singular values rise to match the dominant radial mode at the critical temperature, producing d-fold degeneracy at T_c. This degeneracy swap precedes the susceptibility peak, constituting a parameter-free leading indicator of criticality. On two systems with first-order transitions—the 2D q=5 Potts model (weak first-order, $\xi \approx 2500$) and q=10 (strong first-order, $\xi \approx 5$)—no swap occurs. The ratio σ_2/σ_1 at T_c maps monotonically to transition strength: 1.000 (continuous, $\xi \to \infty$), 0.956 (continuous, different universality class), 0.691 (weak first-order), 0.374 (strong first-order). The FIM singular value profile thus provides three diagnostic capabilities from a single observable: (1) detection of phase transitions via singular value growth, (2) classification of transition order via the presence or absence of the degeneracy swap, and (3) quantification of first-order transition strength via σ_2/σ_1. All three are parameter-free—requiring no thresholds, fitting windows, or prior knowledge of T_c.

Contents

1 Introduction

1.1 The Problem: Diagnosing Phase Transitions from Correlation Data

Phase transitions are conventionally detected through macroscopic observables: divergences in the susceptibility χ, peaks in the specific heat C, or discontinuities in the order parameter m. All require either prior knowledge of the critical temperature or a dense temperature sweep with post-hoc analysis. The question motivating this work is whether the two-point correlation function—the most fundamental structural observable of a statistical mechanical system—can itself signal the approach to criticality and distinguish the order of the transition, without reference to macroscopic fluctuation measures.

Information-geometric approaches to phase transitions have been explored from several directions. Zanardi and Paunković [Zanardi & Paunković, 2006] showed that the fidelity susceptibility of quantum ground states diverges at quantum phase transitions. Prokopenko et al. [Prokopenko et al., 2011] related scalar Fisher information to order parameters, demonstrating divergence at criticality. Machta et al. [Machta et al., 2013] showed that the Fisher information of the Ising model's parameter space exhibits universal structure near critical points, connecting information geometry to the renormalization group. These works all employ scalar Fisher information or construct the FIM in parameter space (temperature, coupling constants). None examines the full singular value spectrum of a FIM constructed in *physical space* from measured correlation functions.

1.2 The Fisher Information Matrix as a Structural Probe

In a companion paper [Darling, 2026], we validated that the FIM constructed from exponential kernels on graphs measures spatial dimension: gap-based rank returns exact integers on manifolds, the participation ratio converges to the Hausdorff dimension on fractals, and rank is monotonically non-increasing under coarse-graining (consistent with the Data Processing Inequality). That paper established the *geometric regime* of the Fisher diagnostic toolkit.

The present paper investigates the *thermal regime*: what happens when the exponential kernel is replaced by a physical correlation function? The mathematical construction is identical—position-indexed distributions, score vectors, singular value decomposition of the FIM—but the kernel is now the measured spin–spin correlator $G(r, T)$, which encodes the full thermodynamic state of the system.

The transition between these two regimes is sharp and provable. For an exponential kernel on Manhattan distance, the score vectors for orthogonal lattice directions factorize into orthogonal subspaces, producing rank exactly equal to the spatial dimension d. For *any* non-exponential kernel, this factorization breaks: the ratio $f(n-1)/f(n)$ varies with n, coupling the directions and producing rank $> d$. The thermal correlation function is never exactly exponential—Ornstein–Zernike at high T, power-law at T_c, flat below T_c—so additional modes appear. This scope boundary is not a limitation; it is a structural result that delineates what the FIM measures in each regime.

1.3 Summary of Results

We study four lattice systems spanning two universality classes and both continuous and first-order transitions. The principal findings are:

1. **Degeneracy swap at continuous transitions.** The FIM singular value profile undergoes a structural rearrangement—the lattice-direction singular values rise to match the dominant radial mode—at the critical temperature. This occurs in both the 2D Ising ($\nu = 1$) and 3D Ising ($\nu \approx 0.630$) models.

2. **No swap at first-order transitions.** The q=5 and q=10 Potts models show singular value growth but never degeneracy. The finite correlation length prevents the kernel from spreading sufficiently.

3. **SV$_2$/SV$_1$ as quantitative classifier.** The ratio at T_c is not binary but continuous, mapping to transition strength (Table 1).

4. **Leading indicator.** In all four systems, Fisher diagnostic changes precede the susceptibility peak.

5. **Rank transitions.** In 3D Ising and Potts q=5, the gap-based rank itself transitions ($1 \to 4$ or $1 \to 3$) near T_c, providing an additional criticality signal.

Table 1: Headline comparison across all four systems. SV$_2$/SV$_1$ cleanly separates continuous (> 0.95) from first-order (< 0.70) transitions.

System	Transition	Rank Pattern	Swap?	SV$_2$/SV$_1$ at T_c
2D Ising	Continuous	3 always	**Yes**	1.000
3D Ising	Continuous	$1 \to 4$ at T_c	**Yes**	0.956
Potts q=5	1st-order (weak)	$1 \to 3$ at T_c	No	0.691
Potts q=10	1st-order (strong)	1 always	No	0.374

2 Construction

2.1 The Fisher Information Matrix on Graphs

Let $G = (V, E)$ be a connected graph with graph distance $d(v, u)$. For a kernel function f assigning weights based on distance, define position-indexed distributions:

$$p_{v_0}(u) = \frac{f\big(d(v_0, u)\big)}{Z(v_0)}, \quad Z(v_0) = \sum_{u \in V} f\big(d(v_0, u)\big). \tag{1}$$

For center vertex v_0 with neighbors $w_1, \ldots, w_k$, define score vectors:

$$s_j(u) = \log p_{w_j}(u) - \log p_{v_0}(u). \tag{2}$$

The $k \times k$ Fisher Information Matrix is:

$$F_{ij} = \sum_{u \in V} s_i(u) \cdot s_j(u) \cdot p_{v_0}(u). \tag{3}$$

Its singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k$ encode the information-geometric structure of the kernel at vertex v_0. The construction, observables, and geometric-regime validation are detailed in the companion paper [Darling, 2026].

2.2 The Thermal Kernel

We replace the fixed exponential kernel with the measured spin–spin correlation function:

$$p_{v_0}(u; T) \propto \big|G\big(d_{\text{Manhattan}}(v_0, u), T\big)\big|, \tag{4}$$

where $G(r, T) = \langle s_0\, s_r \rangle - \langle s_0 \rangle \langle s_r \rangle$ is the connected two-point correlation function at temperature T (for Ising), or $G(r, T) = \langle \delta(s_0, s_r) \rangle - 1/q$ for the q-state Potts model. The absolute value ensures positivity. Radial averaging is performed over all site pairs at Manhattan distance r. No free parameters are introduced beyond the temperature T.

2.3 The Scope Boundary: Exponential Kernel Factorization

The connection between the two regimes rests on a factorization property of the exponential kernel. For $f(r) = \exp(-r/\sigma)$ on Manhattan distance $d(v_0, u) = |u_x| + |u_y|$ on a 2D lattice, the score vector for shifting the center one step in the y-direction is:

$$s_{\mathrm{right}}(u_x, u_y) = \frac{|u_y| - |u_y - 1|}{\sigma}. \tag{5}$$

This depends *only* on u_y—the x-component cancels because $\exp(a+b)/\exp(a+c) = \exp(b-c)$. Score vectors for orthogonal directions live in orthogonal subspaces, producing rank $= d$ exactly.

For any non-exponential kernel, the ratio $f(n-1)/f(n)$ varies with n, so the score depends on *both* coordinates. On a 2D lattice, this produces rank 3: two lattice directions plus a radial coupling mode. The scope boundary is sharp: exponential kernel $\to$ rank $= d$; non-exponential kernel $\to$ rank $=$ information-geometric complexity of the kernel.

2.4 Observables

Four diagnostics are extracted from the FIM singular values:

Gap-based rank (D_{eff}): $D_{\mathrm{eff}} = \arg\max_i(\sigma_i/\sigma_{i+1})$—the position of the largest relative drop [Gavish & Donoho, 2014, Zhu & Ghodsi, 2006]. Returns integers; stable across $\sigma \geq 2$.

Participation ratio (PR): $\mathrm{PR} = (\sum \sigma_i)^2 / \sum \sigma_i^2$. Continuous; equals 1 for rank-1, k for all-equal.

SV profile shape: The full normalized singular value vector. Degeneracy patterns encode phase information.

Disorder index (η): $\eta = \sigma_{D_{\mathrm{eff}}+1}/\sigma_{D_{\mathrm{eff}}}$. Quantifies local geometric irregularity.

In the thermal regime, the primary observables are the *SV degeneracy pattern* and the *SV_2/SV_1 ratio* at T_c.

3 Computational Methods

3.1 Monte Carlo Simulation

All four systems are simulated using the Wolff cluster algorithm [Wolff, 1989], which eliminates critical slowing down for the Ising model and provides efficient sampling for the Potts model. Hot starts are used for $T > T_c$; cold starts for $T \leq T_c$.

Table 2: System parameters for all four models.

System	Lattice sizes	T_c (exact)	T/T_c range	# T points
2D Ising	64^2, 128^2, 256^2	$2/\ln(1+\sqrt{2}) \approx 2.269$	0.90–1.50	12
3D Ising	16^3, 32^3, 48^3	4.5115 (numerical)	0.90–1.50	12
Potts q=5	64^2, 128^2, 256^2	$1/\ln(1+\sqrt{5}) \approx 0.852$	0.90–1.50	12
Potts q=10	128^2	$1/\ln(1+\sqrt{10}) \approx 0.701$	0.90–1.50	9

Equilibration uses 200–300 Wolff sweeps per temperature. Measurements are performed over 500 configurations per temperature, with 30 Fisher sample vertices per temperature point. Manhattan distance is used throughout.

3.2 Correlation Function Measurement

For the Ising model: $G(r, T) = \langle s_0\, s_r \rangle - \langle s_0 \rangle \langle s_r \rangle$, where $s_i \in \{-1, +1\}$. For the Potts model: $G(r, T) = \langle \delta(s_0, s_r) \rangle - 1/q$, measuring excess same-state probability relative to the random baseline. Correlation lengths ξ are extracted from exponential fits at high T.

3.3 FIM Construction and SVD

The pipeline is identical to the companion paper [Darling, 2026]. For each sampled vertex v_0: compute score vectors for all k neighbors using the thermal kernel (Eq. 4), form the $k \times k$ FIM via Eq. (3), extract singular values via SVD. Results are reported as mean $\pm$ standard deviation over 30 vertex samples.

3.4 Validation Gates

Before trusting Fisher diagnostics, every system validates: (a) susceptibility peak at $T/T_c \approx 1.00$; (b) magnetization transition; (c) χ scaling with system size consistent with known exponents; (d) correlation function decaying at high T with correct ξ scaling. Energy histogram bimodality is additionally verified for first-order transitions (Potts models).

4 Results

4.1 The Degeneracy Swap: Discovery in the 2D Ising Model

The 2D Ising model on an N=128 lattice has four nearest neighbors, producing a 4×4 FIM with four singular values. Table 3 shows the mean normalized SV profile at five representative temperatures, and Figure 1 displays the full evolution.

Table 3: SV profile evolution in the 2D Ising model (N=128, Manhattan distance).

T/T_c	SV_1	SV_2	SV_3	SV_4	Degeneracy
1.500	1.000	0.225	0.225	0.040	SV_2=SV_3
1.100	1.000	0.674	0.674	0.170	SV_2=SV_3 growing
1.020	1.000	1.000	0.344	0.090	$\mathbf{SV_1}$=$\mathbf{SV_2}$ (swap)
1.000	1.000	1.000	0.542	0.079	$\mathbf{SV_1}$=$\mathbf{SV_2}$
0.900	1.000	0.479	0.479	0.223	SV_2=SV_3 restored

The evolution proceeds through three phases:

Phase I—Paramagnetic ($T \gg T_c$): $G(r)$ decays exponentially with short correlation length ξ. The thermal kernel is sharply peaked. SV_1 dominates (radial direction); SV_2=SV_3 are degenerate and small. A large gap separates SV_1 from the pair.

Phase II—Critical approach ($T \to T_c$): As ξ grows, the kernel spreads. SV_2 and SV_3 grow, then *split*—SV_2 rises to match SV_1 while SV_3 falls behind. At T_c: [1.00, 1.00, 0.54, 0.08].

Phase III—Ordered ($T < T_c$): $G(r) \to m^2$ (flat). The kernel becomes nearly uniform; score vectors approach zero plus noise. SV_2=SV_3 degeneracy is restored. The disorder index η peaks at 0.47—the *highest* value at any temperature, reflecting a noise-dominated FIM rather than rank collapse.

The swap occurs between $T/T_c \approx 1.10$ and 1.04 on N=128. The susceptibility peaks at $T/T_c \approx 1.005$. The FIM structure changes *before* the macroscopic system responds. Gap-based rank is 3.00 ± 0.00 at every temperature and lattice size—a structural consequence of the non-exponential thermal kernel on a 2D lattice, as predicted by the scope boundary (Section 2.3).

4.2 Universality: Confirmation in the 3D Ising Model

The 3D Ising model has six nearest neighbors, producing a 6×6 FIM. On the L=32 lattice, the SV profile evolution is shown in Table 4 and Figure 2.

The FIM has a clean block structure at all temperatures: SV_1 (radial mode), $SV_{2,3,4}$ (lattice-direction triple, preserving cubic symmetry), $SV_{5,6}$ (anti-parallel residual pair). At T_c, all three

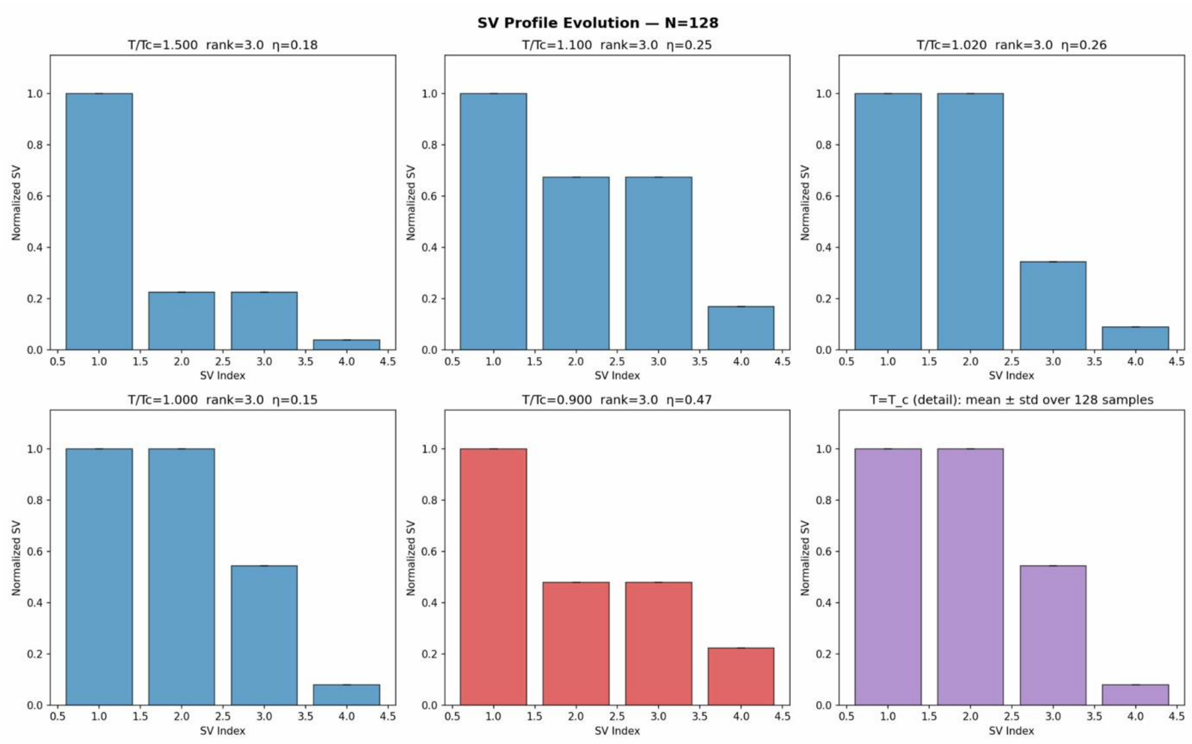

Figure 1: Singular value profile evolution for the 2D Ising model (N=128). Six panels show the FIM SV bar chart at temperatures from $T/T_c = 1.50$ (high T, blue) through $T/T_c = 1.00$ (critical, purple) to $T/T_c = 0.90$ (ordered, red). The degeneracy swap—SV_2=SV_3 at high T transforming to SV_1=SV_2 at T_c—is the signature structural transition. Bottom-right panel shows the mean ± std at T_c. Rank = 3 at all temperatures.

lattice-direction SVs rise *together* to match SV_1, producing quadruple degeneracy. The cubic symmetry is preserved throughout; no sequential or partial swap occurs.

Figure 3 displays the side-by-side comparison at matched T/T_c values, making the dimensional correspondence explicit: in d dimensions, d lattice-direction singular values rise together to match SV_1 at T_c.

4.3 The Rank Transition: A Higher-Dimensional Phenomenon

In the 2D Ising model, gap-based rank is 3 at all temperatures. In the 3D Ising model, a qualitatively new phenomenon appears: the rank *transitions* from 1 to 4 near T_c.

At high T, ξ is very short (0.7–1.4 lattice spacings). The thermal kernel is so sharply peaked that $\mathrm{SV}_{2,3,4} \approx 0.06$ is negligible compared to SV_1. The gap detector places the largest gap between SV_1 and the triple: rank = 1. As $T \to T_c$, the rising triple creates a new largest gap between SV_4 and SV_5: rank = 4.

The transition sharpens with system size (Figure 4, left panel): L=16: rank = 4 by T/T_c = 1.20; L=32: by $T/T_c = 1.04$; L=48: only at $T/T_c = 1.00$. In the thermodynamic limit, the rank transition approaches a step function at T_c—itself a parameter-free criticality diagnostic.

4.4 First-Order Transitions: Potts q=5 and q=10

The 2D q=5 Potts model has a first-order transition at $T_c = 1/\ln(1+\sqrt{5}) \approx 0.852$ [Baxter, 1973], with $\xi \approx 2500$ [Borgs & Janke, 1992]. The q=10 model has $T_c \approx 0.701$ with $\xi \approx 5$ [Borgs & Kotecký, 1990]. Neither shows a degeneracy swap (Table 5).

Table 4: SV profile evolution in the 3D Ising model (L=32).

T/T_c	SV_1	SV_2	SV_3	SV_4	SV_5	SV_6	Rank	Pattern
1.500	1.000	0.059	0.059	0.059	0.028	0.028	1	triple + pair
1.100	1.000	0.198	0.198	0.198	0.079	0.079	1	growing
1.020	1.000	0.663	0.663	0.663	0.185	0.185	4	triple at 66%
1.000	**1.000**	**0.956**	**0.956**	**0.956**	**0.252**	**0.252**	**4**	**quadruple**
0.900	1.000	0.236	0.236	0.236	0.127	0.127	1	triple restored

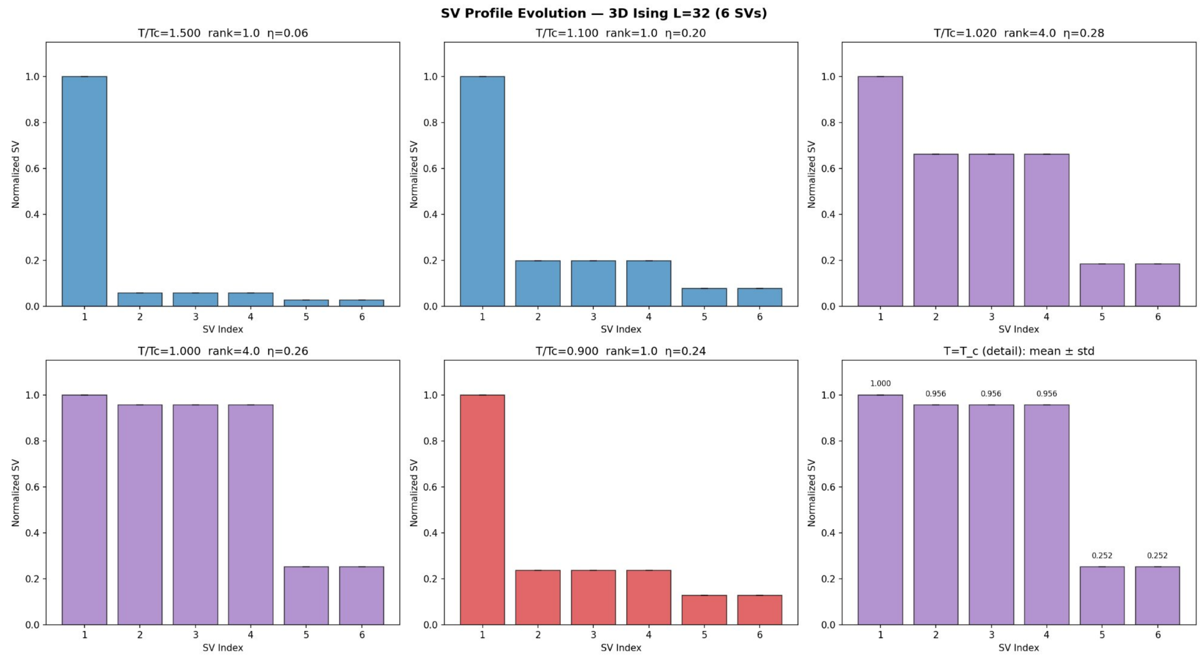

Figure 2: Singular value profile evolution for the 3D Ising model (L=32, 6 SVs). The cubic-symmetry triple $SV_{2,3,4}$ rises from $\sim$0.06 at high T to $\sim$0.96 at T_c, producing quadruple near-degeneracy. The anti-parallel residual pair $SV_{5,6}$ remains small throughout. Rank transitions from 1 to 4.

Figure 5 presents the headline visualization: SV profiles for all three 2D models at five matched temperatures. The contrast between the top row (Ising: swap) and the middle/bottom rows (Potts: no swap) is immediate and unambiguous.

The correlation length at T_c governs the mechanism. Figure 6 confirms that ξ is finite for both Potts models, with q=10 showing a much narrower peak.

Three distinct rank patterns emerge across the 2D models: (1) Ising: rank $=3$ at all T; (2) Potts q=5: rank transitions $1 \to 3$ only at T_c; (3) Potts q=10: rank $=1$ at all T including T_c.

4.5 SV_2/SV_1 as a Quantitative Transition-Order Diagnostic

The ratio SV_2/SV_1 at T_c is not binary but continuous, mapping monotonically to transition strength (Table 6 and Figure 7).

This yields three capabilities from one observable: (1) *detect* whether a transition exists; (2) *classify* continuous vs. first-order (threshold $SV_2/SV_1 \approx 0.80$); (3) *quantify* first-order strength. The gap between continuous (> 0.95) and first-order (< 0.70) is wide. The marginal q=4 Potts model (on the continuous/first-order boundary with logarithmic corrections) is untested and would provide a stringent validation.

Table 5: SV profiles at T_c on 2D lattices (N=128).

Model	Transition	SV_1	SV_2	SV_3	SV_4	SV_2/SV_1	Swap?
2D Ising	Continuous	1.000	1.000	0.542	0.079	**1.000**	**Yes**
Potts q=5	Weak 1st-order	1.000	0.691	0.691	0.373	0.691	No
Potts q=10	Strong 1st-order	1.000	0.374	0.374	0.310	0.374	No

Table 6: SV_2/SV_1 at T_c across all tested systems.

SV_2/SV_1	System	Interpretation
1.000	2D Ising	Continuous ($\xi \to \infty$)
0.956	3D Ising	Continuous, different universality class
0.691	Potts q=5	Weak first-order (ξ finite but large)
0.374	Potts q=10	Strong first-order (ξ small)
~ 0	(no transition)	High-T disordered

4.6 The Leading Indicator

In all four systems, the Fisher disorder index η rises above its high-T baseline before the susceptibility χ peaks (Table 7 and Figure 8).

Table 7: Leading indicator assessment.

System (primary size)	T_{Fisher}	T_χ	Lead $\Delta T/T_c$
2D Ising (N=128)	1.10–1.15	1.005	~0.10–0.13
3D Ising (L=32)	1.200	1.020	~0.18
Potts q=5 (N=128)	1.150	1.000	~0.15
Potts q=10 (N=128)	1.200	1.000	~0.20

Caveat: The η threshold detection uses only 2 high-T baseline points, making reported leads upper bounds. A conservative reading: genuine lead of ~0.04–0.10 across all systems. For first-order transitions, the χ peak is extremely sharp, making the "lead" comparison less meaningful.

4.7 Finite-Size Scaling

Rank. In the 2D Ising model, rank = 3 at all T and N. In 3D Ising, the rank transition sharpens with L (Figure 4). In Potts q=5, the rank transition concentrates at T_c (Figure 9). In q=10, rank = 1 everywhere.

Gap ratio. The clearest scaling signal. At high T on 2D Ising, grows with system size ($2.45 \to 4.44 \to 8.20$ at $T/T_c = 1.5$). Collapses to 1.0 near T_c; the collapse point converges toward T_c with increasing N.

η **behavior.** In 2D Ising (constant rank), η shows a minimum at T_c: 0.164 (N=64), 0.147 (N=128), 0.260 (N=256). In 3D, η is complicated by the rank transition—when rank jumps, η switches what it measures. The gap ratio SV_1/SV_2 (dropping $17.0 \to 1.05 \to 4.24$) is the correct rank-independent signal in 3D.

4.8 Below T_c: The Noise-Dominated Regime

The pre-registered prediction (P2-5) expected rank collapse below T_c. The actual behavior is the opposite: rank remains at its thermal-regime value, and η *rises* to its highest values (~0.46

in 2D Ising, ∼0.24 in 3D Ising, ∼0.30 in Potts). Below T_c, $G(r) \approx m^2$; the kernel becomes nearly uniform; score vectors ≈ 0 plus noise. The noise is approximately equal in all directions, making all SVs comparable. This is a *noise-dominated* FIM, not a degenerate one.

4.9 Full Diagnostic Data: 2D Ising (N=128)

Table 8 provides the complete Fisher diagnostics. Figure 10 displays the four-panel diagnostic view.

Table 8: Full Fisher diagnostics, 2D Ising, N=128, Manhattan distance.

T/T_c	Rank	η	PR	Gap Ratio	χ	C
1.500	3.00	0.179	2.016	4.44	1.1	26.88
1.300	3.00	0.176	2.658	2.46	1.2	54.49
1.200	3.00	0.129	2.643	2.39	1.7	109.09
1.150	3.00	0.210	2.437	3.05	6.3	146.75
1.100	3.00	0.253	3.273	1.48	27.0	216.16
1.070	3.00	0.203	3.365	1.06	18.7	282.00
1.040	3.00	0.307	2.974	1.00	79.1	282.07
1.020	3.00	0.261	2.786	1.00	268.5	280.35
1.010	3.00	0.262	2.797	1.00	617.9	355.26
1.005	3.00	0.282	2.845	1.00	838.3	347.28
1.000	3.00	0.147	2.988	1.00	374.7	2.77
0.900	3.00	0.465	3.152	2.09	1.0	0.78

5 Discussion

5.1 The Mechanism: Why the Swap Occurs at Continuous Transitions

At a continuous transition, $\xi \to \infty$. The thermal kernel spreads across the entire lattice. When the kernel is wide relative to the lattice spacing, moving the center by one site in *any* direction produces approximately the same fractional change in the distribution—all directions contribute equally to the FIM. This equalizes the lattice-direction SVs with SV_1.

At a first-order transition, ξ remains finite. The kernel never fully spreads. Moving radially always produces a larger information change than moving tangentially. $\mathrm{SV}_1 > \mathrm{SV}_2$ persists.

The swap threshold corresponds to ξ/L becoming $O(1)$. For continuous transitions, this occurs at T_c in the thermodynamic limit. For first-order transitions, it never occurs. The 3D Ising result ($\mathrm{SV}_2/\mathrm{SV}_1 = 0.956$) likely reflects the smaller ξ/L achievable on L=32, since ξ diverges more slowly ($\nu \approx 0.630$ vs. $\nu = 1$ in 2D).

5.2 Relation to Prior Work

The FIM has been used as a probe of phase transitions in several contexts, but always as a *scalar* quantity or a matrix in *parameter* space. Zanardi and Paunkovć [Zanardi & Paunkovć, 2006] used fidelity susceptibility (ground state overlaps). Prokopenko et al. [Prokopenko et al., 2011] related scalar Fisher information to order parameters. Machta et al. [Machta et al., 2013] studied the FIM in model parameter space. Transtrum et al. [Transtrum et al., 2015] connected information geometry to effective theories via the sloppy model framework.

Our construction differs in three respects: (1) the FIM is built in *physical space*, not parameter space; (2) the kernel is the *measured* correlation function; (3) the diagnostic is the full *singular value spectrum*, not a scalar. The degeneracy pattern is invisible to scalar Fisher information measures.

5.3 The Two-Regime Framework

The companion paper [Darling, 2026] established the geometric regime (exponential kernel → dimension estimator). This paper establishes the thermal regime (correlation kernel → structural probe). The scope boundary connects them: exponential kernel → rank $= d$; non-exponential kernel → rank encodes kernel complexity. The same mathematical object does different useful things in different regimes.

5.4 Limitations and Open Questions

1. **Limited model family.** Only nearest-neighbor models on regular lattices tested.
2. **The q=4 boundary.** The Potts model at the continuous/first-order boundary is untested.
3. **Leading indicator threshold.** The detection uses an ad hoc 2σ threshold over a 2-point baseline.
4. **Radial averaging.** Per-site correlations might reveal additional structure.
5. **Four data points.** The $\mathrm{SV}_2/\mathrm{SV}_1$ gradient rests on four systems.
6. **Continuous symmetry.** The XY model ($O(2)$ symmetry) is untested.

6 Conclusion

We have shown that the Fisher Information Matrix constructed from measured spin–spin correlation functions exhibits a spectral phase transition—the degeneracy swap—that is universal across continuous transitions (tested in two universality classes) and absent at first-order transitions (tested at two strengths).

The $\mathrm{SV}_2/\mathrm{SV}_1$ ratio at T_c provides a quantitative diagnostic for transition order: values above 0.95 indicate continuous transitions; values below 0.70 indicate first-order. The FIM structural changes precede macroscopic fluctuation observables in all four systems.

These findings establish the FIM singular value spectrum as a parameter-free toolkit for diagnosing phase transitions directly from correlation data, complementing the geometric-regime dimension estimation in the companion paper [Darling, 2026].

Future Directions

Three directions are immediate: the q=4 Potts boundary test; application to financial correlation matrices; and the XY model for universality beyond discrete symmetry groups.

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Table 9: Pre-registered prediction outcomes. Score: 6 PASS, 1 PARTIAL, 7 FAIL.

Phase	ID	Prediction	Result	Reinterpretation
2	P2-1	SV step→graded by T/T_c=1.05	PARTIAL	Swap, not step→graded
2	P2-2	$\eta(T_c) > 0.45$	FAIL	η *minimum* at T_c
2	P2-4	PR negative slope at T_c	PASS	—
2	P2-5	$D_{\text{eff}}(0.9 \cdot T_c) < 2$	FAIL	Noise-dominated, not collapse
2	P2-6	Fisher leads susceptibility	PASS	—
2	P2-7	$\text{PR}(T_c) \in [1.6, 2.1]$	FAIL	PR≈2.75–2.99
3D-1	P3D-1	Rank = 4 at all T	FAIL	Rank transition $1 \to 4$
3D-1	P3D-2	Degeneracy swap occurs	PASS	—
3D-1	P3D-3	η minimum at T_c	FAIL	Rank transition artifact
3D-1	P3D-4	Fisher leads susceptibility	PASS	—
3D-2	P3D2-1	No swap at q=5	PASS	—
3D-2	P3D2-2	No swap at q=10	PASS	—
3D-2	P3D2-4	Rank = 3 at all T (q=5)	FAIL	Rank transition $1 \to 3$
3D-2	P3D2-5	Fisher leads χ (q=5)	PASS	—

A Pre-Registered Predictions and Outcomes

The seven failures cluster into: (a) geometric-regime expectations applied to thermal-regime data (P2-2, P2-5, P2-7), and (b) rank predictions assuming constant rank (P3D-1, P3D-3, P3D2-4). Two failures produced findings more interesting than the original predictions.

B 2D vs. 3D Ising Comparison

Table 10: Side-by-side comparison of 2D and 3D Ising Fisher diagnostics.

Observable	2D Ising	3D Ising
Universality class	ν=1, γ=7/4	ν≈0.630, γ≈1.237
FIM size	4×4	6×6
Rank (high T)	3	1
Rank (at T_c)	3	4
High-T degeneracy	SV_2=SV_3 (pair)	$SV_{2,3,4}$ (triple)
At-T_c degeneracy	SV_1=SV_2	SV_1≈$SV_{2,3,4}$
Swap type	One SV rises	All d SVs rise together
SV_2/SV_1 at T_c	1.000	0.956
η at T_c	0.147	0.263
η below T_c	0.465	0.236
Swap observed?	**Yes**	**Yes**
Leading indicator	Yes ($\Delta T/T_c$≈0.13)	Yes ($\Delta T/T_c$≈0.18)
Rank transition?	No	Yes ($1 \to 4$)

C Supplementary Figures

D Computational Details

All code is Python (NumPy, SciPy [Virtanen et al., 2020], Matplotlib), running on an M4 MacBook Air (24 GB RAM). Full source: `https://github.com/IanD25/D_eff-Test-1.0`.

Table 11: Runtime summary.

Experiment	Total runtime	Per T point
Phase 2: 2D Ising (3 sizes)	∼10 min	∼50 s
Phase 3D-1: 3D Ising (3 sizes)	∼15 min	∼75 s
Phase 3D-2: Potts q=5 + q=10	∼12 min	∼10–37 s
Total	**∼37 min**	

Table 12: Ground truth reference values.

System	T_c	Transition	Source
2D Ising	$2/\ln(1+\sqrt{2}) \approx 2.269$	Continuous	Onsager [Onsager, 1944]
3D Ising	4.5115	Continuous	Ferrenberg & Landau [Ferrenberg & Landau, 1991]
Potts q=5	$1/\ln(1+\sqrt{5}) \approx 0.852$	First-order	Baxter [Baxter, 1973]
Potts q=10	$1/\ln(1+\sqrt{10}) \approx 0.701$	First-order	Baxter [Baxter, 1973]

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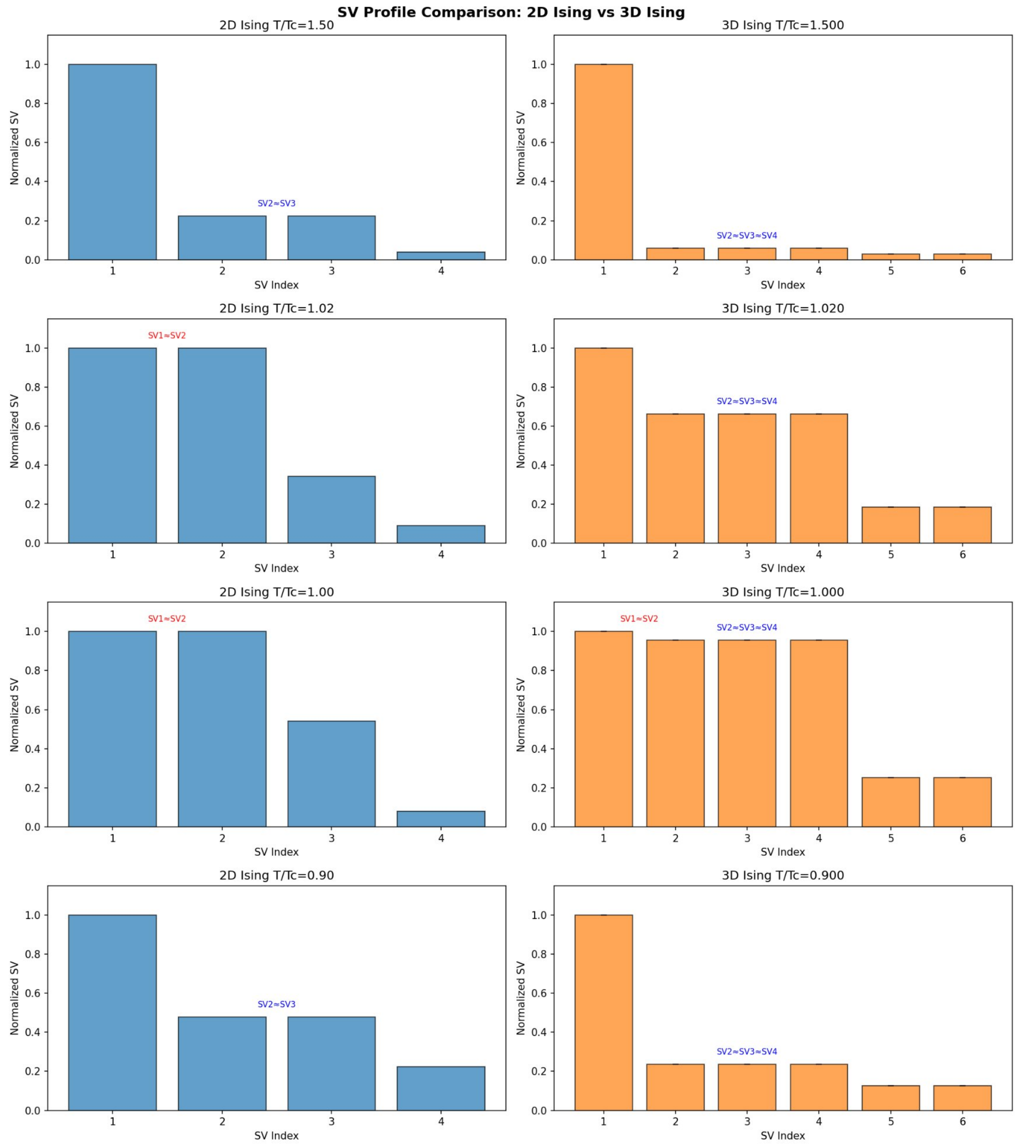

Figure 3: Side-by-side SV profiles for 2D Ising (left, 4 SVs) and 3D Ising (right, 6 SVs) at five matched T/T_c values. In 2D, one lattice-direction SV rises to match SV_1; in 3D, three rise together. The number of rising SVs equals the spatial dimension d.

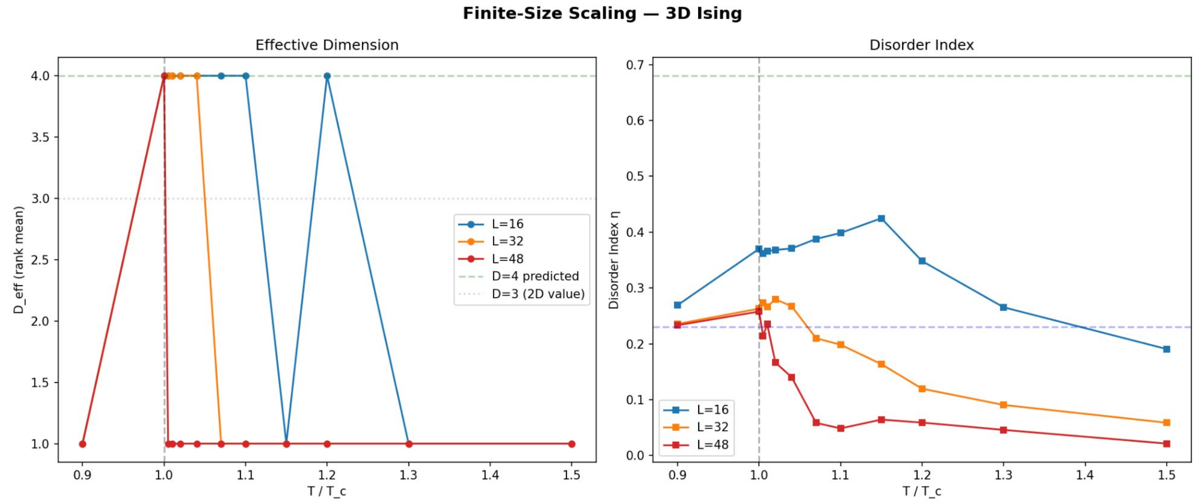

Figure 4: Finite-size scaling for the 3D Ising model. **Left:** Effective dimension vs. T/T_c for $L = 16$, 32, 48. The rank transition sharpens with system size. **Right:** Disorder index η vs. T/T_c. Reference lines show Phase 1 benchmarks.

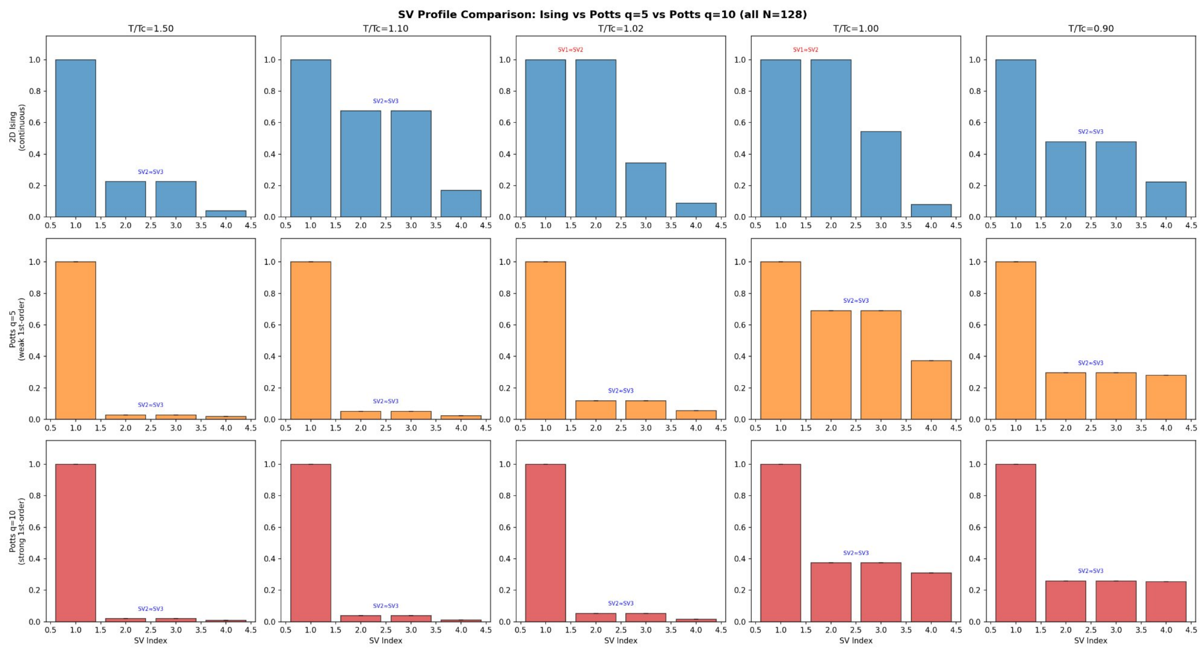

Figure 5: **The central result.** SV profile comparison across three 2D models at five matched temperatures (all N=128). **Top:** 2D Ising (continuous)—SV_1=SV_2 at $T/T_c = 1.02$ and 1.00. **Middle:** Potts q=5 (weak first-order)—SV_2 grows to 0.691 but never matches SV_1. **Bottom:** Potts q=10 (strong first-order)—SV_2 reaches only 0.374. The swap is present at continuous transitions and absent at first-order transitions.

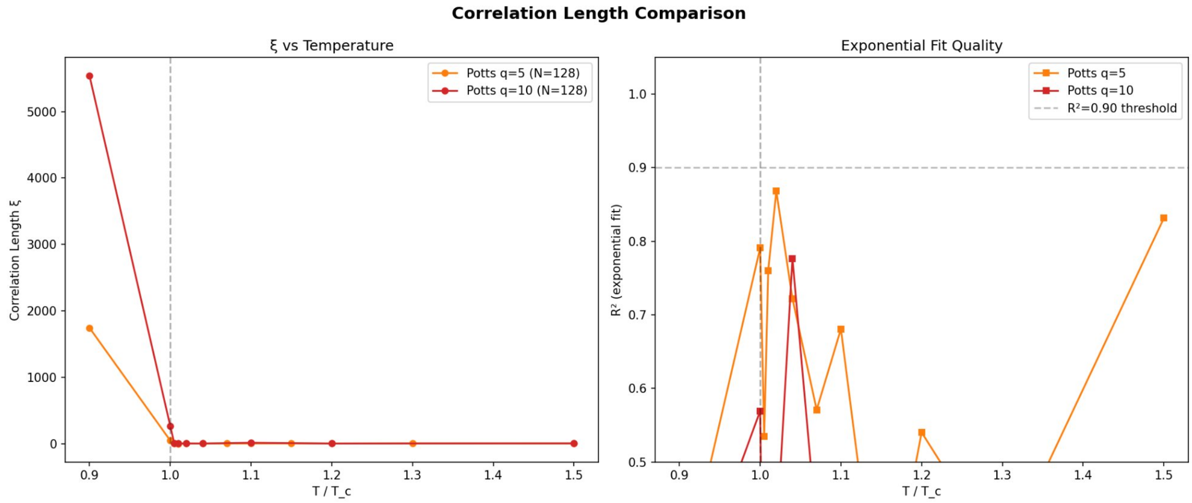

Figure 6: Correlation length and exponential fit quality for the Potts models. **Left:** ξ vs. T/T_c—finite at all T for both models. **Right:** R^2 of exponential fit drops below 0.90 near T_c, indicating departure from pure exponential decay.

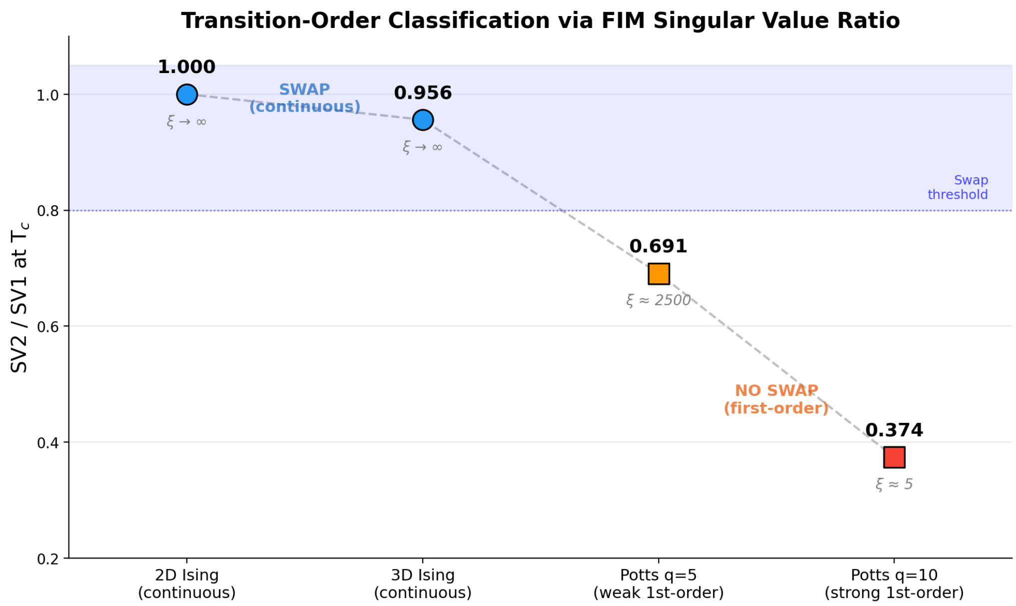

Figure 7: SV_2/SV_1 at T_c as a function of transition type. Continuous transitions cluster above 0.95; first-order transitions fall below 0.70. The gap is wide, and the ratio maps monotonically to the correlation length at T_c. The blue zone indicates the empirical swap threshold.

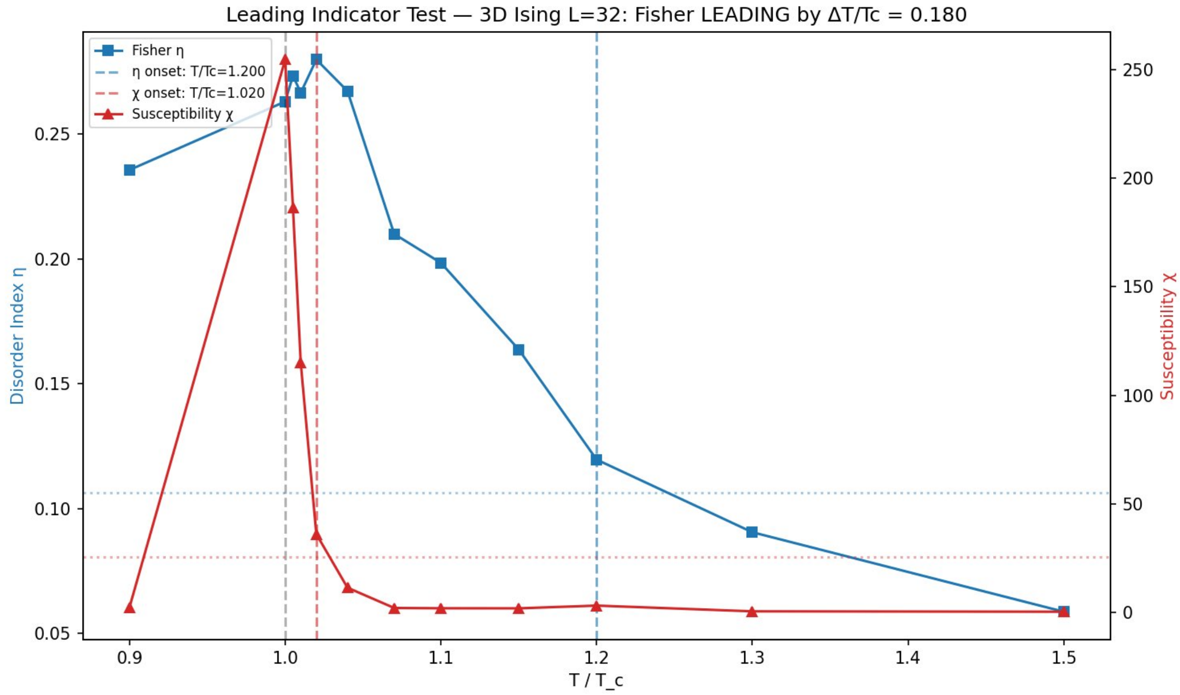

Figure 8: Leading indicator test for the 3D Ising model (L=32). Blue: Fisher η vs. T/T_c. Red: susceptibility χ (right axis). The η onset (T/T_c=1.20, dashed cyan) precedes the χ onset (T/T_c=1.02, dashed red) by $\Delta T/T_c \approx 0.18$.

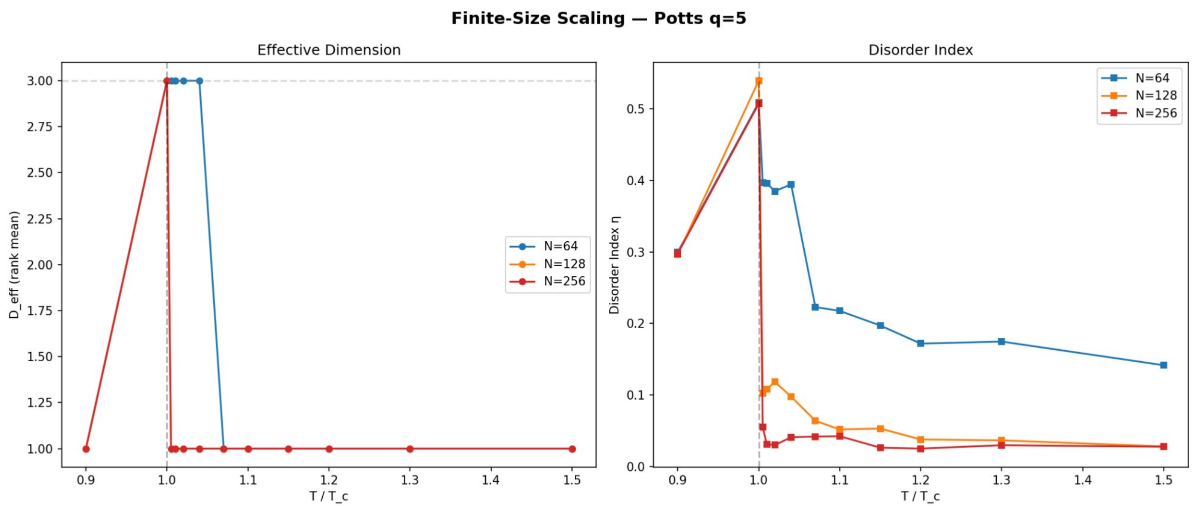

Figure 9: Finite-size scaling for Potts q=5. **Left:** Rank vs. T/T_c—the transition sharpens with N. **Right:** Disorder index η peaks sharply at T_c, narrowing with N.

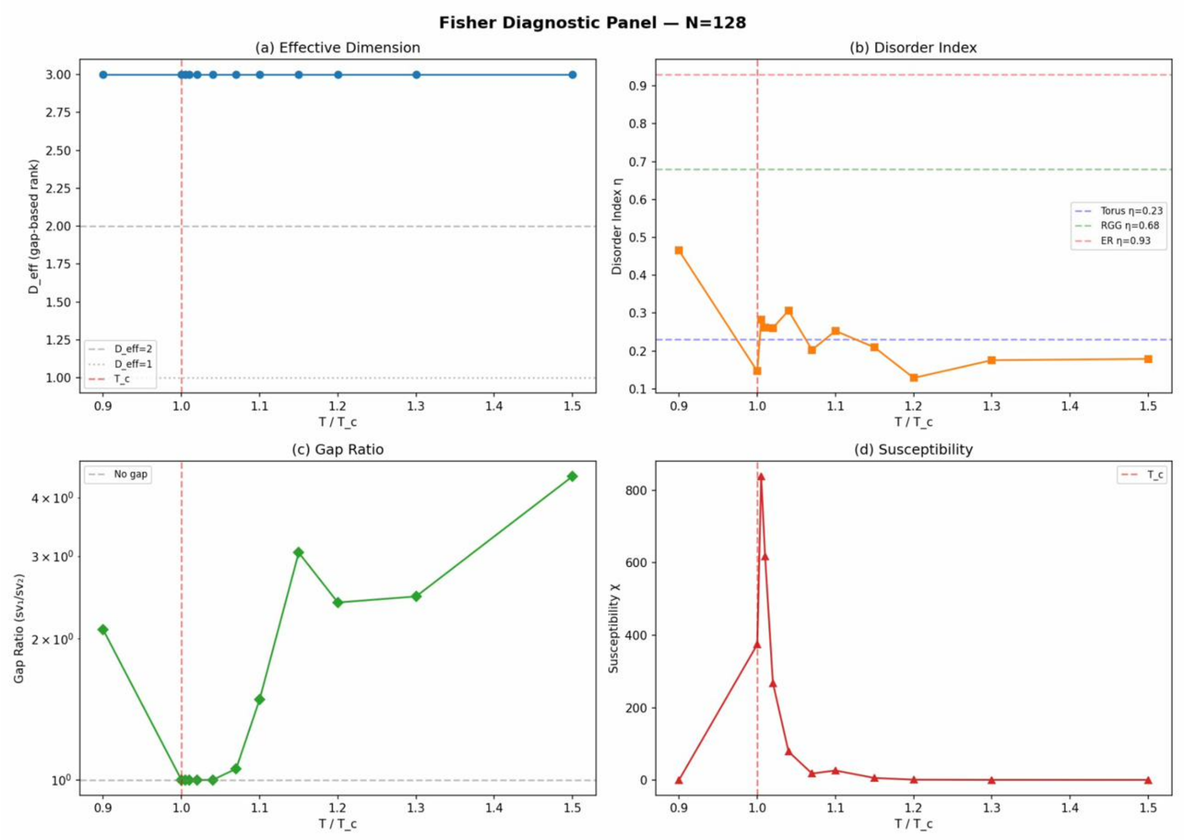

Figure 10: Fisher diagnostic panel for the 2D Ising model (N=128). (a) Effective dimension (rank = 3 at all T). (b) Disorder index η with Phase 1 reference lines. (c) Gap ratio, collapsing to 1.0 at T_c. (d) Susceptibility χ peaking near T_c.

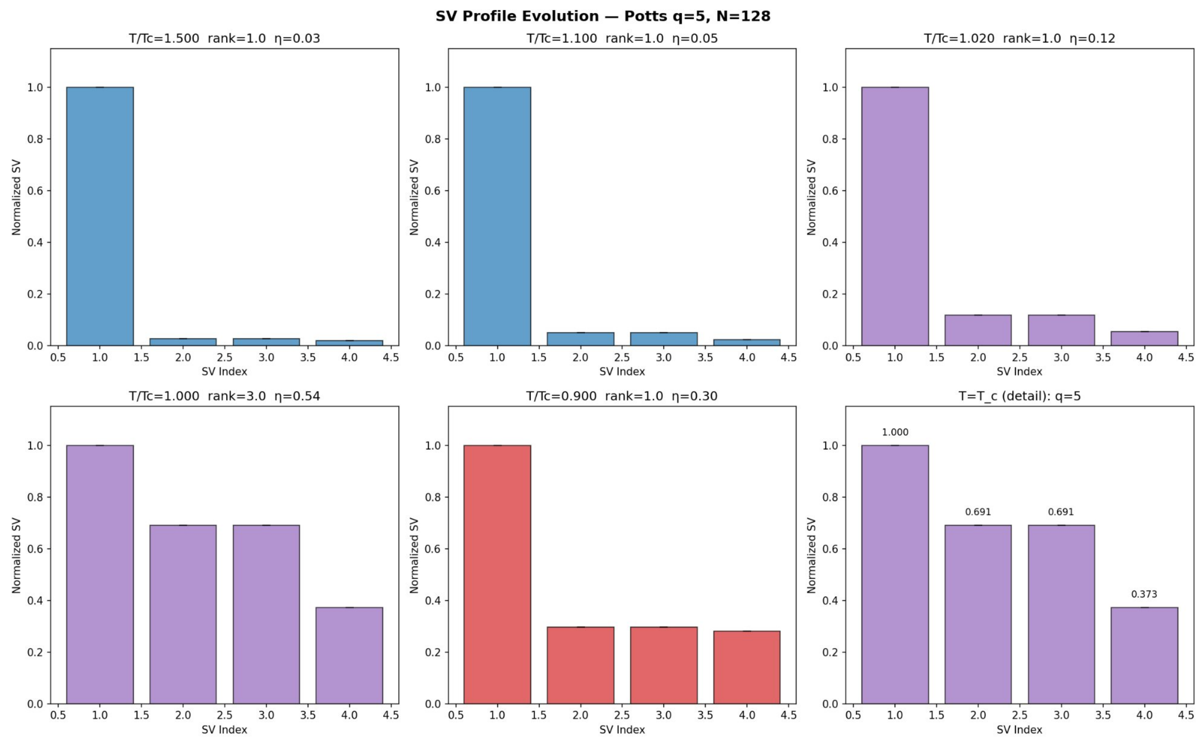

Figure 11: SV profile evolution for Potts q=5 (N=128). Rank transitions 1 → 3 only at T_c; SV_2/SV_1=0.691—no swap.

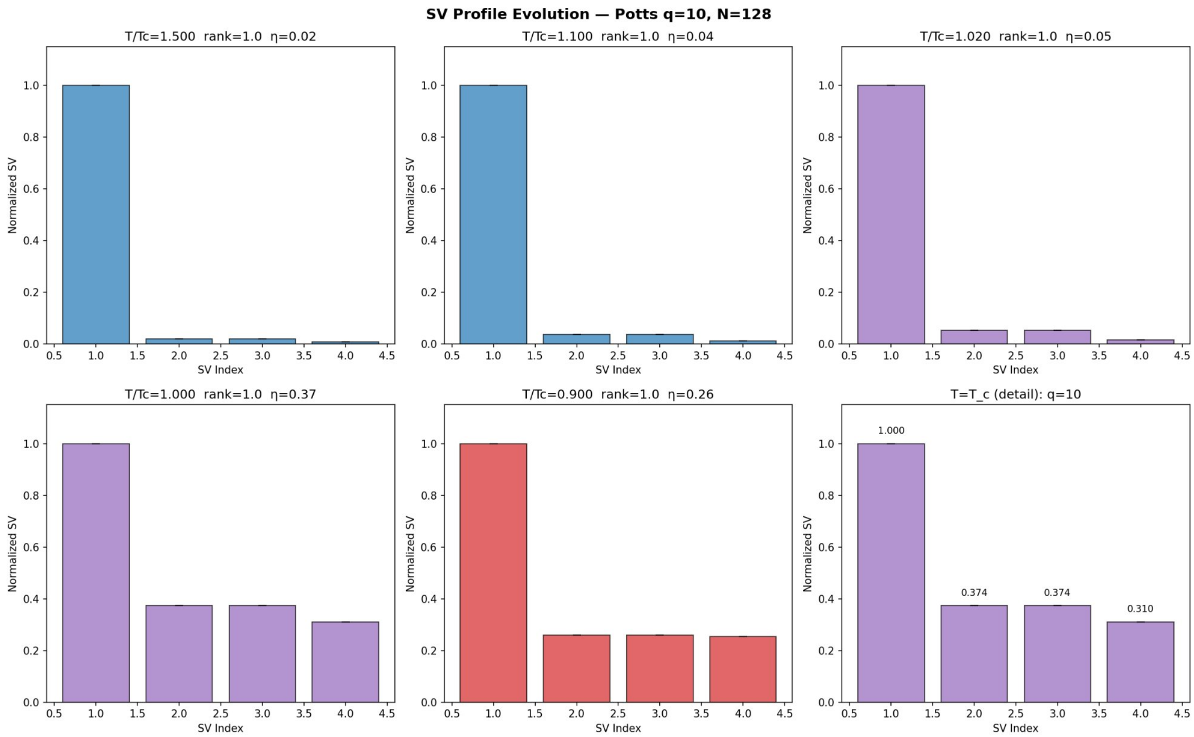

Figure 12: SV profile evolution for Potts q=10 (N=128). Rank = 1 at all T; SV_2/SV_1=0.374—no swap.

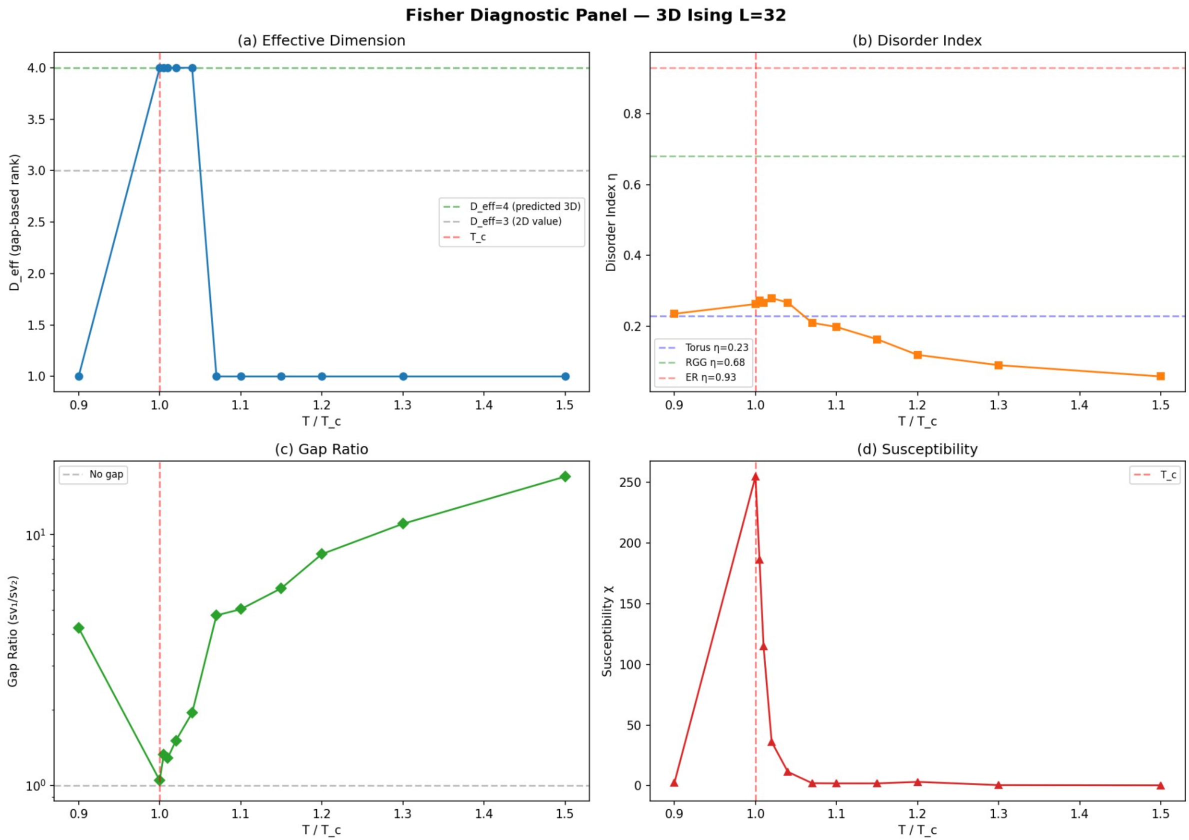

Figure 13: Fisher diagnostic panel for 3D Ising (L=32). (a) Rank transition $1 \to 4$. (b) η peaks near T_c. (c) Gap ratio collapse. (d) Susceptibility.

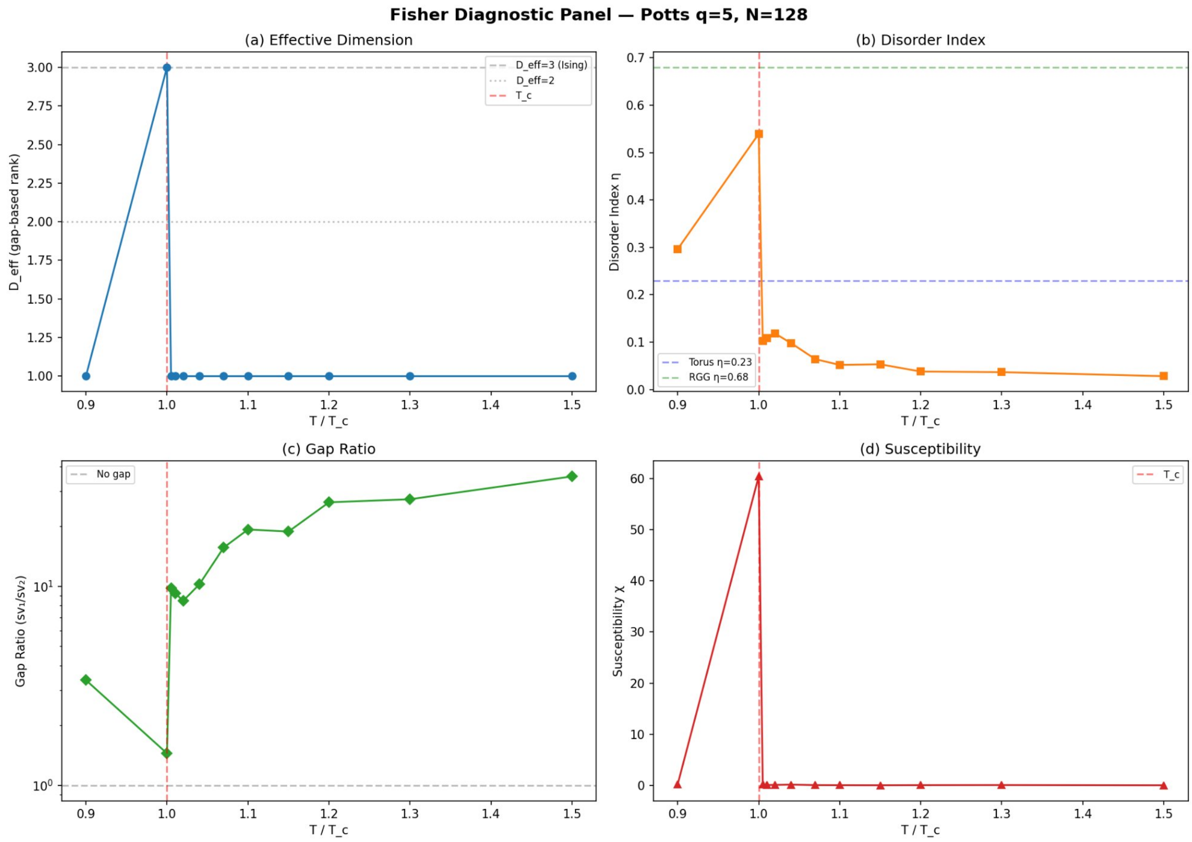

Figure 14: Fisher diagnostic panel for Potts q=5 (N=128). (a) Rank 1 $\rightarrow$ 3 at T_c. (b) η spike. (c) Gap ratio minimum. (d) Sharp χ peak (first-order).

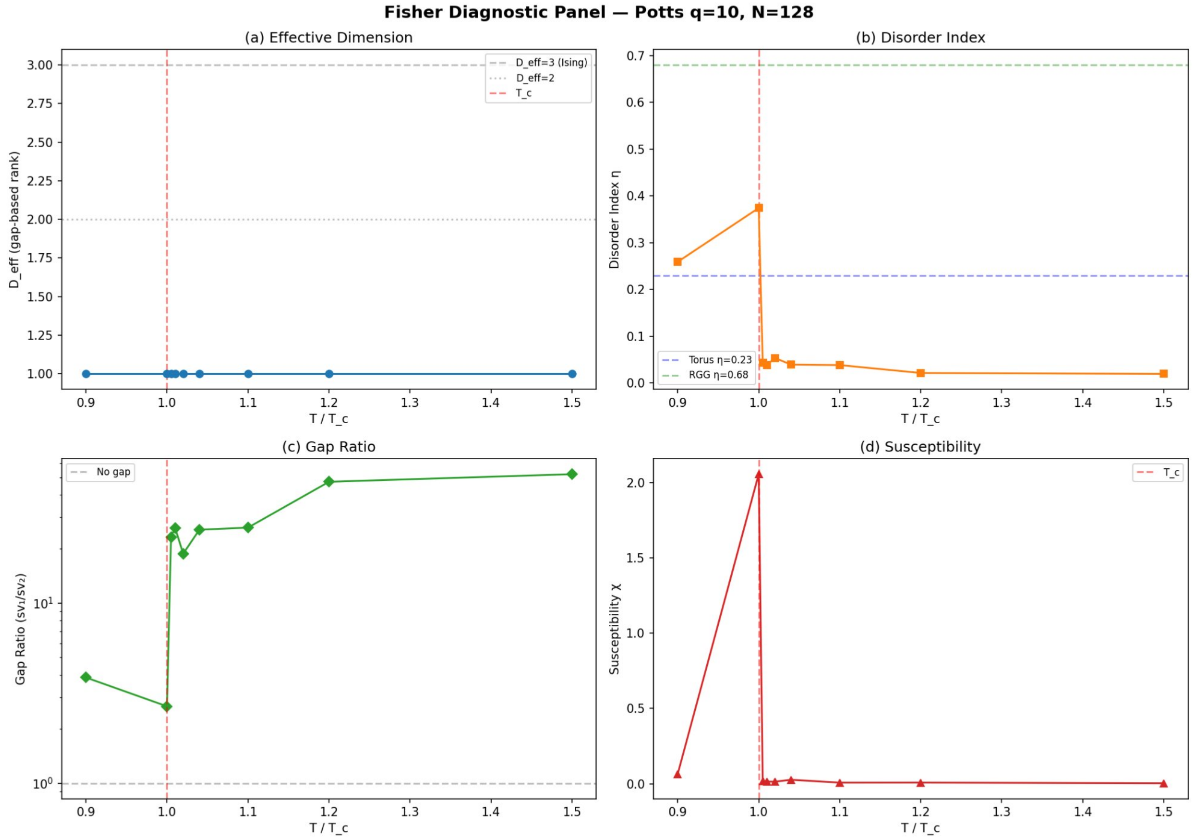

Figure 15: Fisher diagnostic panel for Potts q=10 (N=128). Rank = 1 everywhere; η peaks at T_c but stays below torus reference.

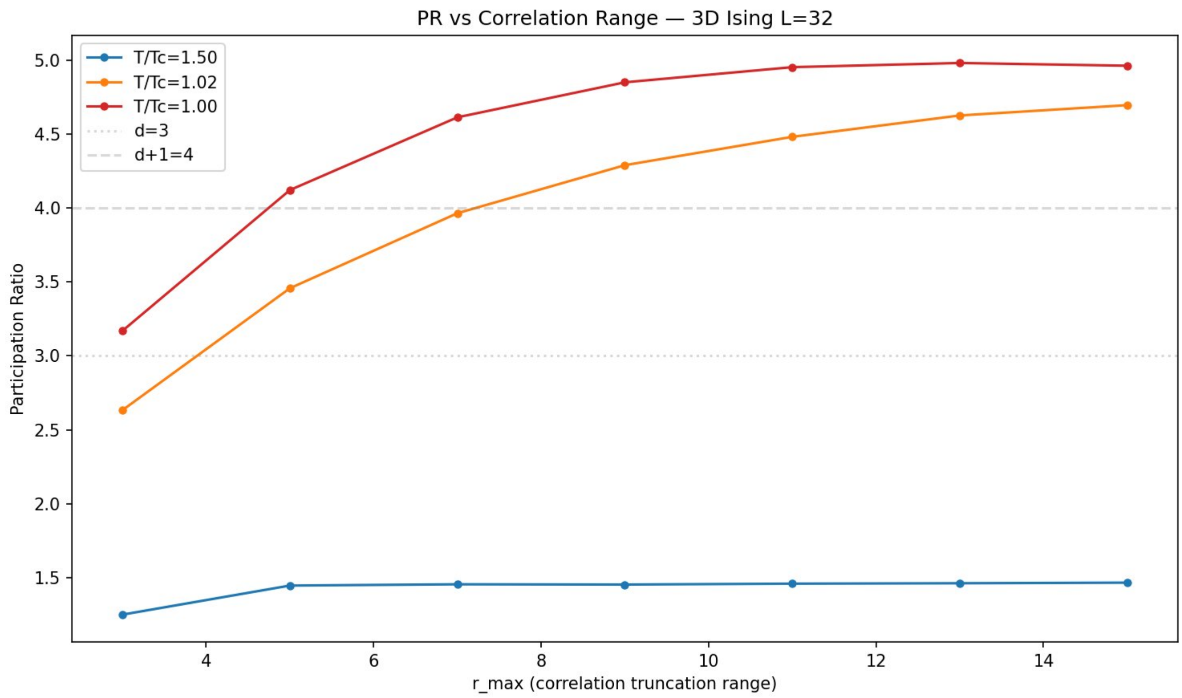

Figure 16: Participation ratio vs. correlation truncation range for 3D Ising (L=32). At high T: PR≈1.4, flat. At T_c: PR exceeds 4 even at small $r_{\max}$.